

A Holistic Approach for Planning Natural Gas and Electricity Distribution Networks

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Abstract—This paper presents a new planning expansion model of an electricity and natural gas distribution system that has high penetration of distributed generation based on natural gas. The model entails lower investment costs as compared to traditional planning models that consider both systems (electricity and natural gas) as independent networks. This model is especially convenient for utilities that own both systems, as electricity and natural gas customers can derive benefits (via electricity or natural gas tariffs) if there are significant savings in investment costs. For fast-growing economies, the merging of electricity and natural gas distribution systems is seen as a promising business opportunity in which economies of scope can have lower investment costs, so there is a definite need for this type of approach.

Index Terms—Distributed generation, electricity distribution system planning, integration of natural gas and distribution systems, natural gas distribution system planning.

NOMENCLATURE

$\sigma_{NS_{i,s,t}}$	Binary decision variable to install a new electrical substation at electrical node i , type s , at the beginning of stage t .
$\sigma_{NF_{i,j,f,t}}$	Binary decision variable to install a new distribution feeder between nodes i - j , type f , at the beginning of stage t .
$\sigma_{NC_{n,c,t}}$	Binary decision variable to install a new city gate, at node n , type c , at the beginning of stage t .
$\sigma_{NP_{n,m,p,t}}$	Binary decision variable to install a new pipeline between nodes n - m , type p , at the beginning of stage t .
$\sigma_{CS_{i,s,t}}$	Binary decision variable to expand the capacity of an existing electrical substation at node i , type s , at the beginning of stage t .
$\sigma_{CF_{i,j,f,t}}$	Binary decision variable to expand the capacity of an existing feeder between nodes i - j , type f , at the beginning of stage t .

$\sigma_{CC_{n,c,t}}$	Binary decision variable to expand the capacity of an existing city gate, at node n , type c , at the beginning of stage t .
$\sigma_{CP_{n,m,p,t}}$	Binary decision variable to expand the capacity of an existing pipeline between nodes n - m , type p , at the beginning of stage t .
$\sigma_{NG_{i,g,t}}$	Binary decision variable to install a new distributed generator at electrical node i , type g , at the beginning of stage t .
$\sigma_{CG_{i,g,t}}$	Binary decision variable to expand the capacity of an existing distributed generator at electrical node i , type g , at the beginning of stage t .
$C_{NS_{i,s,t}}$	Overnight cost of a new electrical substation at node i , type s , at the beginning of stage t (\$).
$C_{NF_{i,j,f,t}}$	Overnight cost of a new distribution feeder between electrical nodes i - j , type f , at the beginning of stage t (\$).
$C_{NC_{n,c,t}}$	Overnight cost of a new city gate, at node n , type c , at the beginning of stage t (\$).
$C_{NP_{n,m,d,t}}$	Overnight cost of a new pipeline, at nodes n - m , type d , at the beginning of stage t (\$).
$C_{CS_{i,s,t}}$	Overnight cost to expand the capacity of an existing electrical substation in electrical node i , type s , at the beginning of stage t (\$).
$C_{CF_{i,j,f,t}}$	Overnight cost to expand the capacity of an existing feeder in electrical nodes i - j , to type f , at the beginning of stage t (\$).
$C_{CC_{n,c,t}}$	Overnight cost to expand the capacity of an existing city gate at node n , type c , at the beginning of stage t (\$).
$C_{CP_{n,m,d,t}}$	Overnight cost to expand the capacity of an existing pipeline that connects nodes n and m , type d , at the beginning of stage t (\$).
$C_{NG_{i,g,t}}$	Overnight cost to install a new distributed generator at node i , type g , at the beginning of stage t (\$).
$C_{CG_{i,g,t}}$	Overnight cost to expand the capacity of an existing distributed generator at node i , type g , at the beginning of stage t (\$).
β	Discount factor.
AP_t	Annual average price of electricity over the stage t (\$/MWh).
LD_l	Total number of hours of the load level l .

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nT	Stage duration in years.	Ω_F	Set formed by new and existing feeders.
nS	Number of stages.	Ω_{NF}	Set formed by new feeders.
nL	Total number of load level demand.	Ω_{CF}	Set formed by existing feeders.
$SS_{i,l,t}$	Power injected by a substation for a load level demand l , at node i , at stage t (MVA).	Ω_C	Set formed by new and existing city gates.
$SG_{i,l,t}$	Power injected by a distributed generator for a load level demand l , at node i , at stage t (MVA).	Ω_{NC}	Set formed by new city gates.
$SD_{i,l,t}$	Forecasted load level demand l , at node i , at stage t (MVA).	Ω_{CC}	Set formed by existing city gates.
$V_{i,l,t}$	Bus voltage for a load level demand l , at node i , at stage t (kV).	Ω_P	Set formed by new and existing pipelines.
$R_{ij,f,t}$	Feeder resistance from node i to j , type f , at stage t (Ohm).	Ω_{NP}	Set formed by new pipelines.
$Z_{ij,f,t}$	Feeder impedance from node i to j , type f , at stage t (Ohm).	Ω_{CP}	Set formed by existing pipelines.
$V_i^{\min}$	Minimum voltage limit at node i (kV).	Ω_n	Set formed by gas nodes that have physical connection with gas node n .
$V_i^{\max}$	Maximum voltage limit at node i (kV).	Ω_i	Set formed by electrical nodes that have physical connection with electrical node i .
$S_{Gg}^{\max}$	Distributed generator type g maximum capacity limit (MVA).	Ω_{GD}	Set formed by gas nodes that have distributed generators.
$S_{Ss}^{\max}$	Electrical substation type s maximum capacity limit (MVA).	Ω_{TS}	Set formed by types of the electrical substation.
$I_f^{\max}$	Type f maximum current limit capacity (kA).	Ω_{TC}	Set formed by types of the city gate.
$\Psi_{Cn,l,t}$	Natural gas injected for the load level demand l , at city gate located at node n , at stage t (m ³ /h).	Ω_{TF}	Set formed by types of the feeders.
$\Psi_{Dn,l,t}$	Forecasted natural gas load level demand l , at node n , at stage t (m ³ /h).	Ω_{TP}	Set formed by types of the pipeline.
$\Psi_{Gn,l,t}$	Natural gas consumption for a load level demand l , of a distributed generator located at gas node n , at stage t (m ³ /h).	Ω_{TG}	Set formed by types of the distributed generator.
$\Psi_{Pn,m,l,t}$	Natural gas flow for the load level demand l , in pipeline between gas nodes $n-m$ at stage t .		
$P_{n,l,t}$	Natural gas pressure for the load level demand l at node n , at stage t (bar).		
$P_{in_{n,m,l,t}}$	Natural gas pressure for the load level demand l , at the input node of the pipeline that connects nodes $n-m$, at stage t (bar).		
$P_{out_{n,m,l,t}}$	Natural gas pressure for the load level demand l , at the output node of the pipeline that connects nodes $n-m$, at stage t (bar).		
$P_n^{\min}$	Minimum natural gas pressure at node n (bar).		
$P_n^{\max}$	Maximum natural gas pressure at node n (bar).		
$\Psi_{Cc}^{\max}$	Type c city gate capacity limit (m ³ /h).		
$\kappa_{nm,p}$	Pipe resistance of the pipeline between nodes $n-m$, type p .		
Ω_E	Set formed by electrical nodes.		
Ω_G	Set formed by natural gas nodes.		
Ω_S	Set formed by new and existing electrical substation.		
Ω_{NS}	Set formed by new electrical substation.		
Ω_{CS}	Set formed by existing electrical substation.		

I. INTRODUCTION

THE most recent findings of natural gas and the low cost of extraction, transportation, and marketing in some developing economies have opened the possibility of incorporating distributed generation (DG) based on natural gas into electricity distribution networks. In fact, an important increment of the new capacity provided by DG in fast-growing economies is expected to come in the next few years [1]. The benefits provided by DG based on natural gas are well known and have been pointed out in several papers [2]–[28], for instance: improvements of voltage regulation and reliability indexes, loss reduction, greenhouse gas reduction, etc. Most of these benefits stem from the fact that DG using natural gas provides firm energy to the system.

The fast penetration and the benefits derived from DG have changed the way electricity utilities operate and plan their distribution systems. The impact of DG on electricity distribution systems has been analyzed from different perspectives; for instance, [3] shows how DG can be taken into consideration in order to reduce energy losses. Likewise, the optimal localization and the capacity of DG to improve voltage profiles are shown in [4] and [5]. The optimal localizations needed to improve voltage profiles and reliability indexes and to reduce energy losses are in [6]–[10]. A multi-objective optimization problem is shown in [11] in order to locate and establish the capacity of different DGs for the purpose of reducing total investment cost. Reference [12] presents an algorithm to locate DGs based on unconventional technologies that mainly consider demand uncertainty. Reference [15] presents an economic and technical analysis of an electrical distribution system that has high penetration of DG based on natural gas.

In [16], a particle swarm optimization algorithm is used to plan a distribution system with DGs so that the investment, operational and CO₂ emission costs are minimized. The algorithm establishes the size and location of DG under different demand and energy price scenarios. On the other hand, [17] presents a model to plan a distribution system in which the investment and operational costs and unsupplied energy risk are considered. The paper uses a Monte Carlo simulation to deal with uncertainties from demand, non-renewable operation of DG, and energy prices. In [18], the methodology presented takes into account the effect of DG on power losses and system operation. This paper considers the uncertainty of ignoring the location and size of DG because it considers the fact that they are investors' decisions that are not within the control of the network operator. In [19], a possibilistic-probabilistic model is presented to plan a distribution system that has different sources of uncertainty. The paper uses different models (fuzzy sets and probability density functions) to model the uncertainty sources. References [20] and [21] show a mathematical model to optimize the operation of multi-carrier energy systems. They use one optimization algorithm to simultaneously minimize the operational cost of the three different systems: electrical, natural gas, and district heat. A model that shows how radial constraints can be easily incorporated in an optimization model of distribution systems is presented in [22]. A distribution-planning optimization model is presented in [23] and [24]. This paper considers a multi-level load, multi-stage planning and the possibility of having DG. The model also considers the possibility of repowering or building new elements, and develops a disjunctive formulation to solve the problem.

On the other hand, several technical papers analyze the problems of planning and operating an integrated electricity power and natural gas system from different perspectives. A long-term, multi-area, multistage, and integrated expansion planning model of electricity and natural gas systems is presented in [25]. This model shows how an electricity and natural gas network can be integrated at the transmission level. Reference [26] presents an integrated transmission-operation planning model of an electricity transmission and a natural gas network. The model emphasizes the need for integrated operation planning in countries that are developing their natural gas infrastructure, as the existing natural gas networks might not be robust enough to supply natural-gas generation plants. In [27], an integrated operation model is solved using a mix of several optimization techniques to speed up the convergence process. Finally, the impact of a natural gas contract on a risk-constrained hydrothermal scheduling problem is presented in [28].

All the previous works that consider DGs at distribution level model DGs as a power injection to the system for operational and planning purposes. In the case of DG based on natural gas, this approach might be practical if a *natural gas network* is designed to supply the gas required for the generator; however, it is often the case that such networks are not designed to supply such a demand—they are often designed to supply residential users or small industrial consumers exclusively. Hence, DG that is modeled as a power injection is inconvenient because it does not take into account the impact of a new natural gas demand (often of considerable size) on the gas network. In other words, distribution planning algorithms that consider DG as a power injection

do not consider the possibility and necessity of redesigning a natural gas network (building and increasing the capacity of new pipelines, city gates, etc.) so as to prevent excessive pressure drops in the natural gas system.

This work proposes a new planning model that integrates electricity distribution and natural gas networks—that is, it considers the two networks as one system. Integrated planning overcomes the many problems and difficulties that a natural gas network might face if a new DG installed in the electricity network requires excessive gas. Integrated planning might achieve lower investment and operational costs for both systems. To the best of our knowledge, there is no such model in the specialized literature. This approach differs from traditional planning algorithms that focus on minimizing the investment and operation costs of electricity and natural gas distribution networks that are considered as two separate networks.

Integrated planning is also seen by many utilities (which operate in the electricity and natural gas sectors as competitors) as an alternative to offering low-cost service to their customers. In other words, utilities that own both electricity and natural gas networks could reduce their investment costs using integrated planning, hence transferring potential savings to their customers via electricity or gas tariffs. In the specific case of Colombia, merging electricity distribution and natural gas companies is seen as an alternative to offering low-cost service [29]. This trend necessarily implies integrated planning. The merging of electricity and natural gas networks has also been observed in other countries, as indicated by a recent report issued by the United Nations [30], which confirms the necessity of having new tools to plan such systems. Finally, integrated planning is also of great importance for countries that are developing their electricity and natural gas infrastructures [31].

This paper is organized as follows. A description of the problem and its mathematical formulation are presented in Sections II and III, respectively. Section IV describes how the new planning model is solved. Numerical results are presented in Section V, and conclusions in Section VI.

II. PROBLEM DESCRIPTION

Electricity distribution planning often focuses on minimizing investment and operational costs over the planning period. The investment cost comprises the cost of building new substations and primary feeders as well as the cost of increasing the capacity of existing substations and primary feeders. Operational costs, on the other hand, include the cost of energy losses over the planning period. The planning is often developed in different stages in order to reduce over cost incurred by the utilities as a consequence of inaccurate demand forecasting. In mathematical terms, the planning model is a mixed-integer nonlinear programming problem.

Natural gas distribution focuses on minimizing the investment costs incurred by utilities installing new pipelines and city gates. The investment cost also includes the cost of increasing the capacity of them. Operational costs are not taken into consideration for natural gas distribution networks, as there are no gas compressor stations at the distribution level; the natural gas required by a compressor station to increase pressure through the system is considered a technical loss. Because gas com-

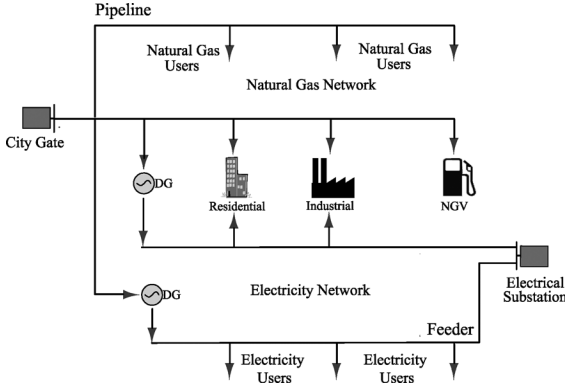


Fig. 1. Integrated electricity and natural gas distribution networks.

pressor stations are located at transmission level, the cost associated with their operation is not considered in the formulation. Similarly, the cost of building natural gas storage facilities and liquefied natural gas plants are not considered at the distribution level, because they provide a better cost/benefit solution at the transmission level [32]. Finally, natural gas distribution may also be developed in stages as an electricity distribution network in order to reduce over cost due to inaccurate natural gas demand forecasting.

For simplicity and illustration purposes, Fig. 1 shows an electricity and a natural gas distribution network. Note how both networks are integrated at different points through a series of various electricity and natural gas demands. In this work, it is assumed that consumers used both electricity and natural gas and that there is a high penetration of natural gas DG as is now occurring in some countries that have low cost of extraction and transportation of natural gas. Fig. 1 shows how DG can be integrated into both networks at any point.

Both networks can be planned to supply future power and natural gas demands in several ways. For instance, the system can be planned as two different networks: that is, the electricity and natural gas distribution systems can be seen as two independent networks in which the natural gas DGs are power injections to the electricity network and demands on the natural gas network. This approach considers DG (power injection or demand) as an exogenous variable to the problem. A different approach is to consider the natural gas DG as an endogenous variable, and plan the whole system as one network.

This paper proposes a new planning model for an integrated electricity and natural gas distribution system such as the one shown in Fig. 1. It is assumed that a utility own both systems, as a conflict of interest might arise between different proprietaries if the networks belonged to different utilities. Conceptually, the new planning model minimizes the investment costs of new elements and of increasing the capacity of existing elements for the whole system (electricity and natural gas) subject to operative constraints. The location and capacity of DGs are exogenous variables if they need to be placed at a specific location or having specific capacity. Otherwise, the model considers location and capacity as endogenous variables in which the location and capacity are not specified by the planner. The final planning model is more complex than models that plan the networks as

two separate systems. However, an integrated model might have better cost/benefit solutions.

Finally, the model does not consider renewable DGs such as wind or solar as it is assumed that natural gas DGs are inexpensive to build and operate. This assumption is valid if the country has significant reserves of natural gas and it is inexpensive (as in Colombia). Additionally, uncertainties relating to renewable energies are not considered, as natural gas can be provided as a firm service. It is also assumed that the utility participates in the electrical and natural gas markets so that it has diversified its investment portfolio in two different energy markets. Finally, it is worth mentioning that this paper assumes that the utility can install natural gas DG in its system as long as the country's regulatory framework allows it.

III. MATHEMATICAL FORMULATION

The integrated planning model is formulated as a mixed-integer nonlinear programming problem. Each component of the model is analyzed separately in order to facilitate its presentation.

A. Objective Function

The objective function (OF) is the present value of six terms that are denoted as $C_t^1, \dots, C_t^6$. In general, the first two terms (C_t^1, C_t^2) represent the cost of installing new elements and expanding the capacity of existing elements of the electricity network. The next two terms (C_t^3, C_t^4) represent the cost of installing new elements and expanding the capacity of existing elements of the natural gas network. The fifth term (C_t^5) represents the cost of installing and increasing the capacity of natural gas DG, and the last term (C_t^6) represents the operative cost of the electricity network. The objective function is given by (1):

$$OF = \sum_{t=1}^{nS} \beta^{(t-1)nT} (C_t^1 + C_t^2 + C_t^3 + C_t^4 + C_t^5 + C_t^6). \quad (1)$$

In general terms, the first five terms of (1) have two components that have a double summation each component. The first component corresponds to the installation of new elements; the second one is associated with the expansion of capacity of existing elements. Likewise, the first summation of each component is associated with the location of the element (substation, primary feeder, city gate, pipelines, etc.) in the system, whereas the second summation is associated with its capacity. Take for instance C_t^1 , which is given by (2). In (2), the first component is the investment cost of a new primary feeder, and the second is the cost of expanding the capacity of existing feeders. Note that the first summations of both components are related to the location of a new or existing feeder and the second summation to its capacity. Equations (3), (4), (5) and (6) have the same interpretation as (2):

$$C_t^1 = \underbrace{\sum_{ij \in \Omega_{NF}} \sum_{f \in \Omega_{TF}} (C_{NF_{ij,f,t}} \sigma_{NF_{ij,f,t}})}_{\text{First component}} + \underbrace{\sum_{ij \in \Omega_{CF}} \sum_{f \in \Omega_{TF}} (C_{CF_{ij,f,t}} \sigma_{CF_{ij,f,t}})}_{\text{Second component}}. \quad (2)$$

The second term (C_t^2) given by (3) is the investment cost of a new primary substation and the cost of expanding the capacity of existing substations:

$$C_t^2 = \sum_{i \in \Omega_{NS}} \sum_{s \in \Omega_{TS}} (C_{NS_{i,s,t}} \sigma_{NS_{i,s,t}}) + \sum_{j \in \Omega_{CS}} \sum_{s \in \Omega_{TS}} (C_{CS_{j,s,t}} \sigma_{CS_{j,s,t}}). \quad (3)$$

The third term (C_t^3) is given by (4) and is the investment cost of new pipelines and the cost of expanding the capacity of existing pipelines:

$$C_t^3 = \sum_{nm \in \Omega_{NP}} \sum_{d \in \Omega_{TP}} (C_{NP_{nm,d,t}} \sigma_{NP_{nm,d,t}}) + \sum_{nm \in \Omega_{CP}} \sum_{d \in \Omega_{TP}} (C_{CP_{nm,d,t}} \sigma_{CP_{nm,d,t}}). \quad (4)$$

The fourth term, (C_t^4), is given by (5) and is the investment cost of new city gates and the cost of expanding the capacity of existing city gates:

$$C_t^4 = \sum_{n \in \Omega_{NC}} \sum_{c \in \Omega_{TC}} (C_{NC_{n,c,t}} \sigma_{NC_{n,c,t}}) + \sum_{m \in \Omega_{CC}} \sum_{c \in \Omega_{TC}} (C_{CC_{m,c,t}} \sigma_{CC_{m,c,t}}). \quad (5)$$

The fifth term, (C_t^5), is given by (6) and is the investment cost of a new natural gas DG and the cost of expanding the capacity of existing natural gas DGs:

$$C_t^5 = \sum_{i \in \Omega_E} \sum_{g \in \Omega_{TG}} (C_{NG_{i,g,t}} \sigma_{NG_{i,g,t}}) + \sum_{j \in \Omega_{GD}} \sum_{g \in \Omega_{TG}} (C_{CG_{j,g,t}} \sigma_{CG_{j,g,t}}). \quad (6)$$

The last term is given by (7), and is the operative cost of the electricity network that corresponds to the energy losses over the planning period:

$$C_t^6 = \beta^{nT} \sum_{l=1}^{nL} \sum_{ij \in \Omega_F} \sum_{f \in \Omega_{TF}} 3 * AP_t * LD_l * nT \cdot \cdot \cdot * R_{ij,f,t} * \left[\left| \frac{V_{i,l,t} - V_{j,l,t}}{Z_{ij,f,t}} \right|^2 (\sigma_{NF_{ij,f,t}} + \sigma_{CF_{ij,f,t}}) \right]. \quad (7)$$

It was mentioned that there is no operative cost for natural gas, as it is assumed that the network does not have technical losses (gas leaks) or gas consumption due to the compressor.

B. Optimization Model Constraints

The optimization model has different constraints, which are grouped into three categories. The first and second categories are physical and operational constraints of the electricity and the natural gas networks, respectively. The last group is associated with the operational characteristics of the natural gas DG that link the electricity and the natural gas networks.

For planning models, it is often common to discretize the capacity of each element, with each capacity called a type. For instance, the capacities of electrical substations might be 10 MVA, 20 MVA or 30 MVA. In this case, there exist three types of substations, i.e., type 1—10 MVA, type 2—20 MVA, and type 3—30 MVA.

1) *Physical and Operational Constraints of the Electricity Distribution Network:* The physical constraints of the electricity network are the nodal power balance at all nodes of the network, the maximum current capacity of all feeders, the maximum capacity of all substations, and the maximum capacity of all DGs to be installed in the network [(8), (9), (10) and (11), respectively]. Equation (12) is the operational constraint or voltage limit. Equation (13) limits the maximum number of natural gas DGs that can be located at each node of the system. Equation (14) guarantees that only one type of substation is located at one node, and (15) guarantees that only one type of conductor is installed in each section of the network:

$$S_{S_{i,l,t}} = S_{D_{i,l,t}} - S_{G_{i,l,t}} + \dots + \sum_{j \in \Omega_i} \left(V_{i,l,t} * \left(\sum_{f \in \Omega_{TF}} (\sigma_{NF_{ij,f,t}} + \sigma_{CF_{ij,f,t}}) * \left(\frac{V_{i,l,t} - V_{j,l,t}}{Z_{ij,f,t}} \right)^2 \right) \right) \quad \forall i \in \Omega_E; \forall l \in \{1, 2, \dots, nL\}; \forall t \in \{1, 2, \dots, nS\} \quad (8)$$

$$\left| \frac{V_{i,l,t} - V_{j,l,t}}{Z_{ij,f,t}} \right| \leq \sum_{f \in \Omega_{TF}} I_f^{\max} * (\sigma_{NF_{ij,f,t}} + \sigma_{CF_{ij,f,t}}) \quad \forall ij \in \Omega_F; \forall l \in \{1, 2, \dots, nL\}; \forall t \in \{1, 2, \dots, nS\} \quad (9)$$

$$S_{S_{i,l,t}} \leq \sum_{s \in \Omega_{TS}} S_{S_s}^{\max} * (\sigma_{NF_{i,s,t}} + \sigma_{CF_{i,s,t}}) \quad \forall i \in \Omega_S; \forall l \in \{1, 2, \dots, nL\}; \forall t \in \{1, 2, \dots, nS\} \quad (10)$$

$$S_{G_{i,l,t}} \leq \sum_{g \in \Omega_{TG}} S_{G_g}^{\max} * (\sigma_{NG_{i,g,t}} + \sigma_{CG_{i,g,t}}) \quad \forall i \in \Omega_E; \forall l \in \{1, 2, \dots, nL\}; \forall t \in \{1, 2, \dots, nS\} \quad (11)$$

$$V_i^{\min} \leq V_{i,l,t} \leq V_i^{\max} \quad \forall i \in \Omega_E; \forall l \in \{1, 2, \dots, nL\}; \forall t \in \{1, 2, \dots, nS\} \quad (12)$$

$$\sum_{g \in \Omega_{TG}} (\sigma_{NG_{i,g,t}} + \sigma_{CG_{i,g,t}}) \leq 1 \quad \forall i \in \Omega_E; \forall t \in \{1, 2, \dots, nS\} \quad (13)$$

$$\sum_{s \in \Omega_{TS}} (\sigma_{NS_{i,s,t}} + \sigma_{CS_{i,s,t}}) \leq 1 \quad \forall i \in \Omega_S; \forall t \in \{1, 2, \dots, nS\} \quad (14)$$

$$\sum_{f \in \Omega_{TF}} (\sigma_{NF_{ij,f,t}} + \sigma_{CF_{ij,f,t}}) \leq 1 \quad \forall ij \in \Omega_F; \forall t \in \{1, 2, \dots, nS\}. \quad (15)$$

2) *Physical and Operational Constraints of the Natural Gas Distribution Network:* The physical constraints of the natural gas network are the nodal natural gas balance at all nodes of the network and the natural gas flow through a pipeline [(16) and (17), respectively]. Equation (17) is derived from the Bernoulli equation applied to a natural gas system [32], [33]. Equation (17) is also the pipeline relation (branch equation) for the natural gas network, in which κ_{Znm} is a function of pipeline con-

struction characteristics such as diameter, length, friction factor, height difference between end points of the pipeline, etc. Equations (18) and (19) are the operational constraints that represent the maximum pressure drop along a pipeline and the maximum city gate capacity, respectively. Equation (20) limits the maximum number of pipelines that can be installed in each network section, and (21) guarantees that only one type of city gate is located at one node:

$$\Psi_{C_{n,l,t}} - \Psi_{D_{n,l,t}} - \Psi_{G_{n,l,t}} - \sum_{m \in \Omega_n} \Psi_{P_{nm,l,t}} = 0$$

$$\forall n \in \Omega_G; \forall l \in \{1, 2, \dots, nL\}; \forall t \in \{1, 2, \dots, nS\} \quad (16)$$

$$\Psi_{P_{nm,l,t}} = \left(\sum_{p \in \Omega_{TP}} (\sigma_{NP_{nm,p,t}} + \sigma_{CP_{nm,p,t}}) \cdot \kappa_{nm,p} \right) \cdot \sqrt{\left(P_{in_{nm,l,t}}^2 - P_{out_{nm,l,t}}^2 \right) - \kappa_{z_{nm}}}$$

$$\forall nm \in \Omega_P; \forall l \in \{1, 2, \dots, nL\}; \forall t \in \{1, 2, \dots, nS\} \quad (17)$$

$$\Psi_{C_{n,l,t}} \leq \sum_{c \in \Omega_{TC}} \Psi_{C_c}^{\max} * (\sigma_{NC_{n,c,t}} + \sigma_{CC_{n,c,t}})$$

$$\forall n \in \Omega_G; \forall l \in \{1, 2, \dots, nL\}; \forall t \in \{1, 2, \dots, nS\} \quad (18)$$

$$P_n^{\min} \leq P_{n,l,t} \leq P_n^{\max}$$

$$\forall n \in \Omega_G; \forall l \in \{1, 2, \dots, nL\}; \forall t \in \{1, 2, \dots, nS\} \quad (19)$$

$$\sum_{p \in \Omega_{TP}} (\sigma_{NP_{nm,p,t}} + \sigma_{CP_{nm,p,t}}) \leq 1$$

$$\forall nm \in \Omega_P; \forall t \in \{1, 2, \dots, nS\} \quad (20)$$

$$\sum_{c \in \Omega_{TC}} (\sigma_{NC_{n,c,t}} + \sigma_{CC_{n,c,t}}) \leq 1$$

$$\forall n \in \Omega_G; \forall t \in \{1, 2, \dots, nS\}. \quad (21)$$

3) Operational Constraint of the DG Based on Natural Gas:

The last constraint, given by (22), is the natural gas required by the DG as a function of electric power injected to the electricity network. In (22), $k_{n,g,1}$ and $k_{n,g,2}$ are the generator's cost-curve coefficients. Observe that (22) links the natural gas network with the electricity network from the operational and physical points of view:

$$\Psi_{G_{n,l,t}} = \sum_{g \in \Omega_{TG}} \left(k_{n,g,2} (S_{G_{i,l,t}})^2 + k_{n,g,1} (S_{G_{i,l,t}}) \right) * (\sigma_{NG_{i,g,t}} + \sigma_{CG_{i,g,t}})$$

$$\forall n \in \Omega_{GD}; \forall l \in \{1, 2, \dots, nL\}; \forall t \in \{1, 2, \dots, nS\}. \quad (22)$$

Note that the previous optimization model is a mixed-integer nonlinear programming model. The nonlinearity arises from the operative cost of the electricity network, the nodal power balance, the natural gas flow through the pipeline, and the natural gas required by the DG. The model is a mixed-integer programming model because the decision variables are integers and the power flow, natural gas flow, gas pressure and nodal voltages are continuous variables. Finally, this model does not include the expected life of different assets, as it is often assumed that the planning horizon time is lower than their expected lives.

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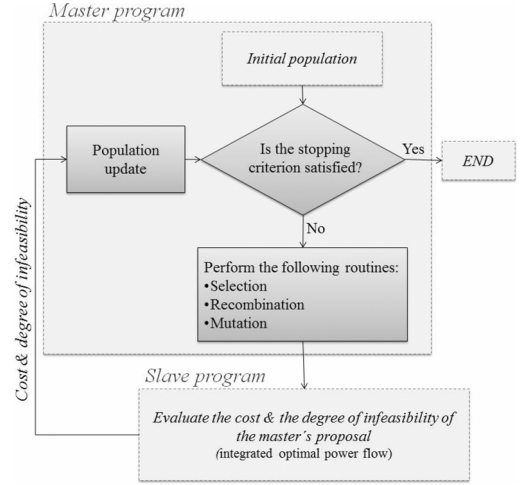


Fig. 2. A simplified flow chart of the master-slave solution technique.

IV. SOLUTION METHODOLOGY

As mentioned in Section III, the optimization model is a mixed-integer nonlinear programming model. These models are challenging; they are often solved using metaheuristic techniques [22], [34]–[39]. This work uses a master/slave methodology as a solution strategy in which the master proposes a solution that is subsequently evaluated by the slave. In terms of the problem, the master proposes an electricity distribution network topology and a natural gas distribution network topology. The master also indicates the types of different elements of both networks, i.e., substation capacities, city gate capacities, etc. Mathematically, the master establishes the values of all integer variables of the optimization problem. Having the master's proposal, the slave can then calculate the objective function given by (1). This master/slave strategy is successful as long as the master is able to propose different topologies that reach lower values of the objective function over the iterative process.

A. Master Program

For this work, a genetic algorithm is used as the master. Specifically, the master program is a Chu-Beasley (CB) genetic algorithm [40] that has been successfully applied in similar problems [34]. Fig. 2 shows the main routines of the master program. These routines are initialization (creation of the initial population), selection, recombination, mutation and population update. In general, they establish different topologies with various types of elements.

The CB genetic algorithm described a system (an electrical and natural gas network topology) as an array in which each component of the array is an integer number that represents a type. Recall that for planning models, the capacity of each element of the system (i.e., electrical and natural gas networks) is a discretized and named type. After the system is represented, the initial population is generated randomly and by a heuristic rule. That is, the rule establishes the topology of the individual step by step based on the minimum distance criteria, i.e., nodes are connected based on the minimum distance between them. On the other hand, element type (capacity) is chosen randomly.

As mentioned before, the CB algorithm performs three main routines, i.e., selection, recombination and mutation, which are shown in Fig. 2. The algorithm uses a tournament as the main operator for selection—that is, two pairs of individuals from the population are selected randomly and the individual (or the system representing an electrical and natural gas network) that has the lowest investment and operational costs for each pair is selected for recombination.

In this work, recombination is implemented based on individual characteristics. An individual characteristic is a segment of the electrical or natural gas network. In an electrical network, a segment might have several electrical feeders, substations, etc., that are physically connected; on the other hand, a segment in a natural gas network might have several pipelines, city gates, etc., that are physically connected. Recombination combines different individual characteristics (in other words, different segments) of the electrical and natural gas networks from the two individuals that were chosen by the selection routines to produce a third system. It is worth mentioning that recombination also takes into account the fact that the result must be a radial system in order to be topologically feasible: hence recombination routines often combine different individual characteristics until a feasible system is found.

Mutation is the last routine. It consists of changing the capacity of one element (its type) or eliminating that element of the system. An integer random number from 0 to the maximum element type is randomly generated. If the number is 0, it implies that the element must be eliminated from the system; otherwise, the number indicates the element type. For instance, an element type is established as 2 if the random number is 2. Note that this routine might create a loop, or it might leave a demand node unconnected to the system. In such a case, a corrective action is implemented: one feeder of the loop is eliminated or the isolated node is connected to the system.

The final routine in Fig. 2 is the population update. The routine might replace the worst individual with the individual that has been evaluated by the slave. The worst individual is defined as the one that has the highest value of objective function or the highest degree of infeasibility in the population (see Section IV-B for an explanation of how this is calculated). The worst individual is substituted by the master's proposal if: 1) it is infeasible but the master's proposal has a lower degree of infeasibility; or 2) it is feasible, but the master's proposal is also feasible and has a lower objective function. Additionally, it is required that the master's proposal have differences in terms of topology and element types for all individuals of the population. Finally, the CB genetic algorithm stops if the incumbent (the system that has the lowest investment and operational costs) has not changed after several iterations.

B. Slave Program

As mentioned before, the slave program evaluates the master's proposal or calculates the objective function of a topology that is found by the CB genetic algorithm. Note that the objective function that is given by (1) has six terms. The first five terms can be easily calculated, as they are functions of the system topology and the capacities of all the elements given by the CB genetic algorithm. The last term, the operational cost

of the electricity network, is needed. This can be calculated by solving an integrated optimal power flow that is derived from the mathematical formulation presented in Section III. That is, an optimal power flow that minimizes the active power losses (P_{LOSS}) subject to the operational constraints of both systems (electricity and natural gas). This integrated optimal power flow is solved for a given topology (given by the master program). When the master proposal (i.e., the integer decision variables) is substituted in all the equations of Section III-B, the mathematical formulation of the integrated optimal power flow is as follows:

$$\min_{\substack{S_{S_{i,l,t}}; S_{G_{i,l,t}}; V_{i,l,t}; \\ \Psi_{C_{n,l,t}}; P_{n,l,t}}} P_{\text{Loss}} = \sum_{i \in \Omega_E} \sum_{l=1}^{nL} \sum_{t=1}^{nS} \text{real} \{S_{S_{i,l,t}} + S_{G_{i,l,t}}\} \quad (23)$$

s.t

$$\begin{aligned} S_{S_{i,l,t}} &= S_{D_{i,l,t}} - S_{G_{i,l,t}} + \sum_{j \in \Omega_i} \left(V_{i,l,t} * \left(\frac{V_{i,l,t} - V_{j,l,t}}{Z_{ij,f,t}} \right)^* \right) \\ \left| \frac{V_{i,l,t} - V_{j,l,t}}{Z_{ij,f,t}} \right| &\leq I_f^{\max} \\ S_{S_{i,l,t}} &\leq S_{S_s}^{\max} \\ S_{G_{i,l,t}} &\leq S_{G_g}^{\max} \\ V_i^{\min} &\leq V_{i,l,t} \leq V_i^{\max} \\ \Psi_{C_{n,l,t}} - \Psi_{D_{n,l,t}} - \Psi_{G_{n,l,t}} - \sum_{m \in \Omega_n} \Psi_{P_{nm,l,t}} &= 0 \\ \Psi_{P_{nm,l,t}} &= \kappa_{nm,p} \sqrt{\left(P_{in_{nm,l,t}}^2 - P_{out_{nm,l,t}}^2 \right)} - \kappa_{zm,n} \\ \Psi_{C_{n,l,t}} &\leq \Psi_{C_c}^{\max} \\ P_n^{\min} &\leq P_{n,l,t} \leq P_n^{\max} \\ \Psi_{G_{n,l,t}} &= k_{n,g,2} (S_{G_{i,l,t}})^2 + k_{n,g,1} (S_{G_{i,l,t}}). \end{aligned} \quad (24)$$

This problem is called integrated optimal power to differentiate it from the classical ac optimal power flow because the natural gas network constraints are incorporated in the formulation. Note that the active power injections of all natural gas DGs are calculated from this formulation as they are the decision variables of the optimization problem. Observe also that the natural gas network constraints are necessary in order to have feasible active power injections to the electricity network that do not violate the operational limits of the natural gas network.

The previous optimization problem is a nonlinear model that might not converge, thus rendering the master proposal (topology) infeasible. In this case, it is possible to relax all violated constraints and solve it again to calculate the degree of infeasibility. Once the degree of infeasibility has been obtained, the individual might be part of the population as explained in the last paragraph of Section IV-A. Finally, if the optimization model converges, the operational cost of the electrical network or power losses can be determined. The fitness function of the master's proposal is the sum of the total investment cost [the first five terms of (1)] and the operational cost that is calculated from the objective function of the integrated optimal power flow [the last term of (1)].

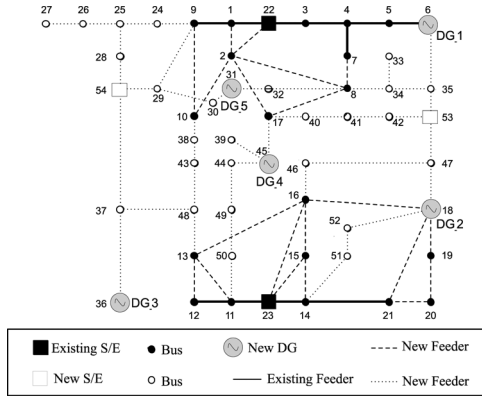


Fig. 3. Electricity distribution network.

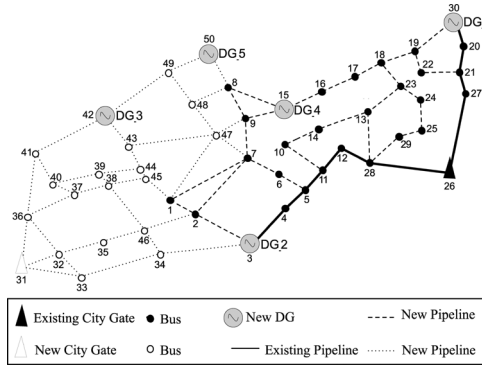


Fig. 4. Natural gas distribution network.

V. NUMERICAL RESULTS

A. System Description

The proposed methodology is applied to an electricity power distribution system and a natural gas distribution system, shown in Figs. 3 and 4, respectively. The planning horizon is ten years, which are divided in two stages of five years each in order to reduce power-demand forecasting errors. It is also common to discretize the load-duration curve when planning a power distribution network. In this work, three load levels are considered, i.e., 100%, 60%, and 30% of peak electrical demand, with durations of 1095, 5475 and 2190 hours, respectively. Average load growth is expected to be 1.5% over the planning period.

In both figures, solid lines represent existing elements at the beginning of the planning period, dashed lines represent potential network expansions for the first stage, and dotted lines represent potential network expansions for the second stage. The filled circles are the electrical nodes (Fig. 3) and natural gas nodes (Fig. 4) that either exist or could be connected in the first stage of the planning period. Unfilled circles are the electrical nodes (Fig. 3) and natural gas nodes (Fig. 4) that will be connected at the second stage. All nodes might have power demand or natural gas demand. If they have demand, they must be connected to the system in the first or second stage. Potential locations of new natural gas DGs are shown as gray circles. Finally, it is important to mention that the system planner identifies these potential expansions as mainly due to the city's growth projections.

The electricity distribution network has 14 existing nodes, 2 existing substations, and 11 feeders. To supply the projected electric power demand, 20 potential new feeders for the first stage, 41 potential new feeders for the second stage, and 2 new substations might be built. On the other hand, the natural gas network has 20 existing nodes, 1 existing city gate, and 10 pipelines. To supply the projected natural gas demand, 27 potential new pipelines for the first stage, 36 new pipelines for second stage, and 1 city gate might be built. Additionally, 5 new natural gas DGs that might be installed in the system. The system at the end of the planning period is expected to be four times as large as the existing system in terms of power demand and number of new elements. Finally, the full system database, including power demand, installation costs, energy costs, etc., is available from the authors upon request.

B. Analysis of Results

The numerical results are obtained from the algorithm described in Section IV. Adaptive mutation and recombination rates are also used to speed up the convergence process. This strategy allows the master program to explore a wide region of the solution space during the first phases of the convergence process and refine the search for a better solution during the last phases of the convergence process. The master program is written in Matlab and the slave program in GAMS (solver CONOPT).

In order to analyze the benefits of the proposed methodology, three different cases are studied. The first case calculates the expansion costs of the electricity and natural gas networks as two independent networks that are not linked by a DG. This case is used to determine expansion cost while ignoring DG penetration. The second case evaluates the expansion cost of having DG penetration as a power injection to the electricity system, and a gas demand to the gas system. This case is a non-integrated expansion plan in which the DG is seen as an exogenous variable to both problems. The last case is the integrated expansion plan that was formulated in Section III and solved using the strategy described in Section IV. These three cases compare the effectiveness of different planning alternatives in terms of the total expansion cost. All cases have the same planning horizon (10 years and 2 stages) and the same multi-level (discretized) load described in Section V.A. Note that cases 2 and 3 model NG-DGs as active/reactive power injections subject to active/reactive power generation limits. They are modeled in such a way because NG-DGs are part of an optimal power flow formulation for both cases. There is, however, a difference between cases 2 and 3: in case 2, DGs are also modeled as a given natural gas demand (an exogenous variable) for the natural gas network, whereas in case 3 the natural gas demand is a decision variable (an endogenous variable) for the OPF.

To solve the integrated multi-stage and multi-load mathematical model, three different metaheuristic techniques have been programmed to compare the performance of the solution approach described in Section IV. A classical genetic algorithm (GA), a classical particle swarm optimization algorithm (PSO) and a Chu-Beasley genetic algorithm (CB) are used. A classical genetic algorithm and a particle swarm optimization algorithm were chosen for this study as they have been reported in other

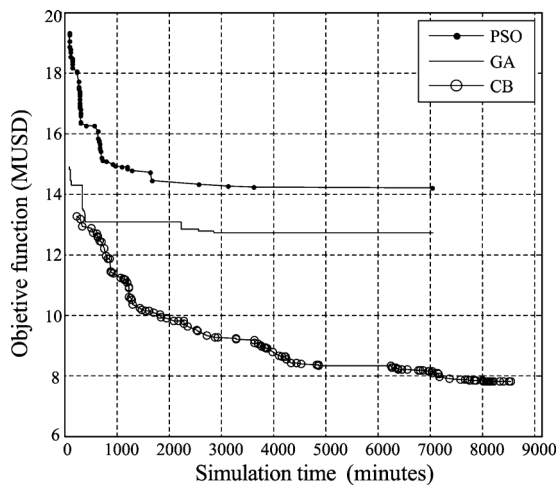


Fig. 5. Comparison of convergence.

TABLE I
TOTAL EXPANSION COST IN MILLIONS OF US DOLLARS

Stage	Description	Case 1	Case 2	Case 3
1	Expansion cost of the electric network	2.49	2.10	2.10
	Operative cost of the electric network	1.99	0.93	0.97
	Expansion cost of the natural gas network	0.47	1.69	0.73
	Stage 1 total cost	4.95	4.72	3.80
2	Expansion cost of the electric network	5.03	4.44	4.48
	Operative cost of the electric network	3.16	1.89	2.27
	Expansion cost of the natural gas network	0.82	1.87	1.04
	Stage 2 total cost	9.01	8.20	7.79
TOTAL COST (Stage 1 + Stage 2)		13.96	12.92	11.59

papers to solve distribution planning models [16], [17], [35], [36] and [37]. Fig. 5 shows that the lowest investment and operational cost system is achieved with the master/slave strategy for case 3, stage 2, which corresponds to the more complex computational case. Note that the Chu-Beasley genetic algorithm that is implemented is able to reduce the fitness function by 38.8% with respect to the second best option, i.e., genetic algorithm. However, Fig. 5 also shows that a master/slave strategy based on a Chu-Beasley genetic algorithm generates better solutions if enough simulations are performed over time.

The consolidated results are shown in Table I. The table shows the present value of the total expansion cost at stages 1 and 2 of the three cases. Note that, as expected, integrated planning (case 3) has the lowest expansion cost, confirming the benefits of the model presented in this paper. The most expensive expansion was found in case 1, which does not have DG penetration, indicating that natural gas DG penetration reduces investment cost, as shown in cases 2 and 3.

Tables II and III break down the expansion costs of the electricity and natural gas systems. Note that the highest expansion cost for the electricity distribution network corresponds to case 1 and the lowest cost to case 2. Observe also that electricity

TABLE II
EXPANSION COST OF ELECTRICITY DISTRIBUTION NETWORK IN MILLIONS OF US DOLLARS

Stage	Description	Case 1	Case 2	Case 3
1	Expansion cost of distribution feeders (new or repowering)	2.09	1.60	1.60
	Expansion cost of substations (new or repowering)	0.40	0.00	0.00
	Cost of New DGs	0.00	0.50	0.50
	Stage 1 total cost	2.49	2.10	2.10
2	Expansion cost of distribution feeders (new or repowering)	4.44	3.33	3.54
	Expansion cost of substations (new or repowering)	0.59	0.42	0.60
	Cost of New DGs	0.00	0.69	0.34
	Stage 2 total cost	5.03	4.44	4.48
TOTAL COST (Stage 1 + Stage 2)		7.52	6.54	6.58

TABLE III
EXPANSION COST OF THE NATURAL GAS DISTRIBUTION NETWORK IN MILLIONS OF US DOLLARS

Stage	Description	Case 1	Case 2	Case 3
1	Expansion cost of pipelines (new or repowering)	0.47	0.76	0.73
	Expansion cost of city gates (new or repowering)	0.00	0.93	0.00
	Stage 1 total cost	0.47	1.69	0.73
2	Expansion cost of pipelines (new or repowering)	0.44	0.95	0.66
	Expansion cost of city gates (new or repowering)	0.38	0.92	0.38
	Stage 2 total cost	0.82	1.87	1.04
TOTAL COST (Stage 1 + Stage 2)		1.29	3.56	1.77

expansion cost of case 2 (USD 9.36 M) is lower than case 3 (USD 9.82 M), as case 3 has a higher operational cost. Even though case 3 has a higher electricity expansion and operative cost than case 2, the expansion cost of the natural gas network in case 3 (USD 1.77 M) is lower than in case 2 (USD 3.56 M), as can be observed in Table III. In fact, the lower cost of natural gas expansion compensates for the higher cost of the electricity network in case 3, so that the integrated expansion model costs less.

On the other hand, Figs. 6–8 show the topology that has the lowest investment cost for *the first planning period or stage 1* for cases 1, 2 and 3, respectively. Additionally, Figs. 9–11 show the topology that has the lowest investment cost for *the second planning period or stage 2* for cases 1, 2 and 3, respectively. The most relevant findings that can be observed from Figs. 6–8 (stage 1) are as follows:

- The presence of DGs in Figs. 7 and 8 eliminates the need to repower some of the power distribution feeders and electrical substations when DG is not considered (Fig. 6). This can be observed from the number above (in parenthesis) the segments that connects nodes 3-4, 15-16 and 22-15 in Figs. 6–8. A lower number inside the parenthesis means a less expensive element type.

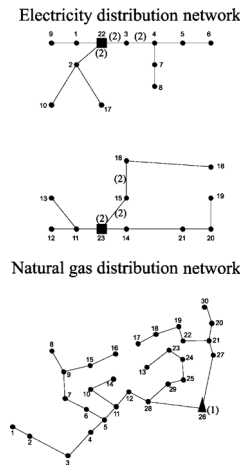


Fig. 6. Case 1—Electricity and natural gas distribution network topologies—Stage 1.

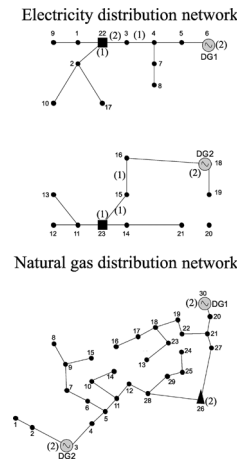


Fig. 7. Case 2—Electricity and natural gas distribution network topologies—Stage 1.

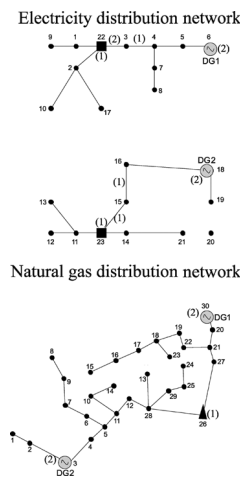


Fig. 8. Case 3—Electricity and natural gas distribution network topologies—Stage 1.

- The presence of DGs also allows end nodes to be supplied in such a way that the operational cost of the network is reduced. This can be observed in node 19, which is connected to the network from node 20 (Fig. 6), whereas it is connected from node 18 in Figs. 7 and 8.

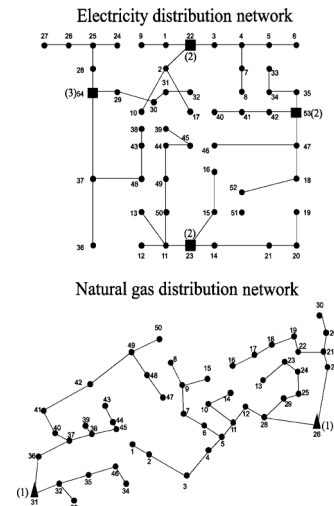


Fig. 9. Case 1—Electricity and natural gas distribution network topologies—Stage 2.

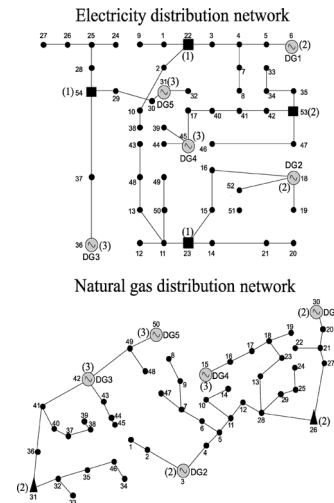


Fig. 10. Case 2—Electricity and natural gas distribution network topologies—Stage 2.

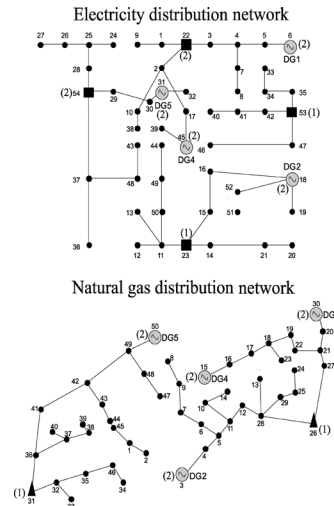


Fig. 11. Case 3—Electricity and natural gas distribution network topologies—Stage 2.

- Observe that cases 2 and 3 have the same electricity distribution network but different natural gas networks (Figs. 7

and 8). From Table I, note that the operational cost of the electricity network is higher and the expansion cost of the natural gas network is lower in case 3 than in case 2. The integrated expansion plan is therefore able to find a combination that has the lowest total cost.

On the other hand, the most relevant findings that can be observed from Figs. 9–11 (stage 2) are listed below. It is worth mentioning that this planning stage is more complex than the first stage, as the electrical and natural gas networks are larger than those planned for the previous stage.

- The algorithm finds that there are economical benefits if some of the electrical power feeders or natural gas pipelines change from closed to open, even though some of these feeders or pipelines were built during the first stage. This can be observed in the electrical feeders that connect 16–18 (Figs. 6 and 9) and pipelines 9–15 and 19–22 (Figs. 7 and 10).
- Note that cases 2 and 3 have different electricity and natural gas distribution networks (Figs. 10 and 11). This situation was different at end of stage 1 (Figs. 7 and 8). The algorithm finds that it is necessary to have different topologies in order to reduce the total cost of case 3. Additionally, the algorithm indicates that the capacity and location of the DGs in Figs. 10 and 11 need to be different. Specifically, the DGs at nodes 31 and 45 have a lower capacity, and node 36 does not have a DG in Fig. 11. These changes make case 3 the best option in terms of total cost.

Finally, note that numerical results of stages 1 and 2 confirm that a lower expansion cost is achieved if the electrical and natural gas networks are planned as one system. That is, the cost/benefit analysis of case 3 (integrated planning) and case 2 (traditional separate expansion planning) shows that a holistic approach is able to find lower-cost solutions.

VI. CONCLUSION

This paper presents a new distribution expansion model that considers electricity distribution and natural gas networks as one system with a high penetration of distributed generation based on natural gas. Numerical results show that a holistic approach reduces the expansion cost of both networks, as compared to a traditional approach in which electricity and natural gas networks are planned as separate systems. The cost reduction is achieved because natural gas DG is an endogenous variable in the holistic approach, whereas it is an exogenous variable (power injection of gas demand) in the classical approach. The proposed method is particularly convenient for utilities that own electricity distribution and natural gas networks, as investment cost reduction can mean significant benefits for electricity and natural gas customers. That is, electricity and natural gas distribution tariffs are often cost-based services that depend on the expansion costs and other factors. Hence, any savings on these activities might benefit customers via electricity or natural gas tariffs. Finally, this methodology is expected to be implemented in Colombia, as various utilities are considering merging their electricity and natural gas businesses, and hence will take advantage of the economies of scope of such a business opportunity.

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